

Introduction to Special Relativity

Chapter 1: The Postulates and the Relativity of Simultaneity

Study Notes Series

1 Why Classical Mechanics Needed a Fix

By the late nineteenth century, physicists had two extremely successful theories: Newtonian mechanics and Maxwell's electromagnetism. The trouble was that they disagreed about something fundamental: the speed of light. Newtonian mechanics assumed velocities simply add together (if you throw a ball forward from a moving train, its speed relative to the ground is the train's speed plus the ball's speed). Maxwell's equations, however, predicted that light travels at a fixed speed $c \approx 3.00 \times 10^8$ m/s regardless of the motion of its source. Experiments, most famously the Michelson-Morley experiment, confirmed that light does not obey the usual addition of velocities.

In 1905, Einstein resolved the contradiction not by patching electromagnetism, but by rewriting the rules of space and time themselves.

2 The Two Postulates

Special relativity rests on two postulates:

1. **Principle of relativity:** The laws of physics take the same form in every inertial (non-accelerating) reference frame.
2. **Constancy of the speed of light:** The speed of light in vacuum, c , has the same value in every inertial frame, independent of the motion of the source or the observer.

These two statements sound modest, but they force a complete rethink of simultaneity, time, and distance.

3 Simultaneity Is Not Absolute

Consider two events that occur at different locations. In Newtonian physics, if two observers agree the events happened "at the same time," every observer agrees, regardless of how they are moving. Special relativity destroys this assumption.

A classic illustration: a light bulb sits at the exact center of a moving train car. When the bulb flashes, light travels toward both the front and back walls of the car.

- An observer sitting inside the train sees the light hit the front and back walls *simultaneously*, since both walls are equidistant from the bulb and light travels at c in both directions.
- An observer standing on the platform, watching the train go by, sees the back wall moving *toward* the point where the light was emitted and the front wall moving *away* from it. Since

light still travels at c in this observer's frame too, the light reaches the back wall first. Both observers are correct. Simultaneity depends on the observer's frame of reference.

4 The Lorentz Transformation

The mathematical machinery that connects the coordinates (x, t) measured in one inertial frame to the coordinates (x', t') measured in another frame, moving at relative velocity v along the x -axis, is the Lorentz transformation:

$$x' = \gamma(x - vt) \tag{1}$$

$$t' = \gamma\left(t - \frac{vx}{c^2}\right) \tag{2}$$

where the Lorentz factor is defined as

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}. \tag{3}$$

Notice that $\gamma \rightarrow 1$ when $v \ll c$, recovering ordinary Newtonian behavior at everyday speeds. Only as v approaches c does γ grow large and relativistic effects become significant. This factor γ will reappear constantly in the next chapter, where it governs time dilation and length contraction.

Quick check

If a spacecraft travels at $v = 0.6c$, what is γ ? (Answer: $\gamma = 1/\sqrt{1 - 0.36} = 1.25$.)